

Assignment 3, ML-Lab

(Model evaluation, Cross validation)

1. Set A3 - Create 'User' Data set having 5 columns namely: User ID, Gender, Age, EstimatedSalary and Purchased. Build a logistic regression model that can predict whether on the given parameter a person will buy a car or not.

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
import metrics

import seaborn as sn

import matplotlib.pyplot as plt

data = pd.read_csv("/home/ctbora/Downloads/suvdata.csv")
x = data.iloc[:, 2:3].values
y = data.iloc[:, 4].values

print(x)
print(y)

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.25, random_state=0)
logistic_regression = LogisticRegression()

logistic_regression.fit(x_train, y_train)
y_pred = logistic_regression.predict(x_test)

confusion_matrix = pd.crosstab(y_test, y_pred, rownames=['Actual'], colnames=['Predicted'])
sn.heatmap(confusion_matrix, annot=True)

print('Accuracy: ', metrics.accuracy_score(y_test, y_pred))
plt.show()

print(x_test)
print(y_pred)
```

Q1. Find the accuracy of a logistic regression model on Iris data by using the K-Fold Cross-Validation.

Q2. Find the accuracy of a logistic regression model on Iris data by using the Stratified K-Fold Cross-Validation.

Q3. Find the accuracy, precision, recall, specificity, ROC, AUC, and F1-score of a logistic regression model on Iris data.